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SECTION : BSE – 4A

DATABASE SYSTEMS
LAB 14

BSSE SPRING 2026

LAB 14

SETUP

SQLQuery1.s...aisal (86)*

```

2  CREATE DATABASE LibraryDB;
3  GO
4
5  -- Use Database
6  USE LibraryDB;
7  GO
8  -- Create Table
9  CREATE TABLE Books (
10     BookID INT PRIMARY KEY,
11     Title VARCHAR(100),
12     AvailableCopies INT
13 );
14 -- Insert Records
15 INSERT INTO Books VALUES
16 (1, 'Database Systems', 5),
17 (2, 'Operating Systems', 3),
18 (3, 'Artificial Intelligence', 4);
19 -- View Data
20 SELECT * FROM Books;
21

```

100 % 1 0 Ln: 2, Ch: 1 SPC CRLF UTF-8

Results Messages

	BookID	Title	AvailableCopies
1	1	Database Systems	5
2	2	Operating Systems	3
3	3	Artificial Intelligence	4

SCENARIO 01

SQLQuery1.s...aisal (86)*

```

1  BEGIN TRANSACTION;
2
3  UPDATE Books
4  SET AvailableCopies = AvailableCopies - 1
5  WHERE BookID = 1;
6
7  -- Cancel Transaction
8  ROLLBACK;
9
10 SELECT * FROM Books;

```

100 % No issues found Ln: 10, Ch: 21 TABS CRLF UTF-8

Results Messages

	BookID	Title	AvailableCopies
1	1	Database Systems	5
2	2	Operating Systems	3
3	3	Artificial Intelligence	4

Query executed successfully. DESKTOP-02Q2MPC\SQLEXPRESS ... DESKTOP-02Q2MPC\Shah F... LibraryDB 00:00:00 Row: 1, Col: 1 3 rows

SCENARIO 02

SQLQuery1.s...aisal (86))*

```

1 BEGIN TRANSACTION;
2
3 UPDATE Books
4 SET AvailableCopies = AvailableCopies + 1
5 WHERE BookID = 2;
6
7 COMMIT;
8
9 SELECT * FROM Books;

```

100 % No issues found Ln: 9, Ch: 21 TABS CRLF UTF-8

Results Messages

	BookID	Title	AvailableCopies
1	1	Database Systems	5
2	2	Operating Systems	4
3	3	Artificial Intelligence	4

SCENARIO 03

SQLQuery1.s...aisal (86))*

```

1 BEGIN TRANSACTION;
2
3 DELETE FROM Books
4 WHERE BookID = 3;
5
6 -- Recover
7 ROLLBACK;
8
9 SELECT * FROM Books;

```

100 % No issues found Ln: 9, Ch: 21 TABS CRLF UTF-8

Results Messages

	BookID	Title	AvailableCopies
1	1	Database Systems	5
2	2	Operating Systems	4
3	3	Artificial Intelligence	4

Query executed successfully. DESKTOP-02Q2MPC\SQLEXPRESS ... DESKTOP-02Q2MPC\Shah F... LibraryDB 00:00:00 Row: 1, Col: 1 3 rows

SCENARIO 04

SQLQuery1.s...aisal (86)*

```

1 BEGIN TRANSACTION;
2 -- Operation 1: Delete
3 DELETE FROM Books WHERE BookID = 1;
4 -- Savepoint
5 SAVE TRANSACTION SavePoint1;
6 -- Operation 2: Update
7 UPDATE Books
8 SET AvailableCopies = AvailableCopies - 1
9 WHERE BookID = 2;
10 -- Undo only second operation
11 ROLLBACK TRANSACTION SavePoint1;
12 -- Save remaining work
13 COMMIT;
14
15 SELECT * FROM Books;

```

100 % No issues found Ln: 1, Ch: 19 TABS CRLF UTF-8

Results Messages

BookID	Title	AvailableCopies	
1	2	Operating Systems	4
2	3	Artificial Intelligence	4

Query executed successfully. DESKTOP-02Q2MPC\SQLEXPRESS ... DESKTOP-02Q2MPC\Shah F... LibraryDB 00:00:00 Row: 1, Col: 1 2 rows

SCENARIO 05

SQLQuery1.s...aisal (86)*

```

1 BEGIN TRANSACTION;
2
3 -- Return Book
4 UPDATE Books
5 SET AvailableCopies = AvailableCopies + 1
6 WHERE BookID = 1;
7
8 -- Borrow Another Book
9 UPDATE Books
10 SET AvailableCopies = AvailableCopies - 1
11 WHERE BookID = 2;
12
13 COMMIT;
14
15 SELECT * FROM Books;

```

100 % No issues found Ln: 15, Ch: 21 TABS CRLF UTF-8

Results Messages

BookID	Title	AvailableCopies	
1	2	Operating Systems	3
2	3	Artificial Intelligence	4

Query executed successfully. DESKTOP-02Q2MPC\SQLEXPRESS ... DESKTOP-02Q2MPC\Shah F... LibraryDB 00:00:00 Row: 1, Col: 1 2 rows

- What happens to the number of copies after cancellation?
The number of copies remains unchanged.
Even though you reduced it during the transaction, the ROLLBACK cancels everything, restoring the original value.
- Which ACID property is applied?
Atomicity
→ The transaction is treated as a single unit: it either completes fully or has no effect at all.
- What is the final number of copies?
The copies increase by 1.
Example: if it was 3, it becomes 4 after the transaction.
- Why is it necessary to make the transaction permanent?
Because without COMMIT, the change is not saved permanently.
If the system crashes or the transaction is rolled back, the update will be lost.
COMMIT ensures Durability — once saved, the data stays even after failures.
- How can the deleted record be recovered?
By using ROLLBACK before committing the transaction.
This cancels the delete operation and restores the record to its previous state.
- What would happen if recovery was not done?
The deletion becomes permanent (especially after COMMIT).
The record is lost, which can lead to data inconsistency and missing information in the system.
- Which operation remains?
The DELETE operation (BookID = 1) remains.
- Which operation is undone?
The UPDATE operation (reducing available copies of BookID = 2) is undone using the SAVEPOINT rollback.
- Why must both operations succeed or fail together?
Because the exchange is a single logical action. If only one operation succeeds (e.g., one book is deducted but the other is not added), the database becomes inconsistent.
Treating both as one unit ensures fairness and data accuracy—either the exchange fully happens or nothing changes.
- Which ACID property is demonstrated?
Atomicity
→ Both operations are executed as a single indivisible unit. Either both succeed together or both are rolled back together.